

Curriculum Vitae

Zahra Mosafer alileh

Email: **zahramosafer99@gmail.com**

Location: Iran

Research Interests

Artificial Intelligence and Scientific Discovery · AI-Augmented Scientific Problem Solving · Human–AI Collaboration · Cognitive Aspects of Problem Solving · Machine Learning · Mathematical Reasoning · Manifold Learning · Dimensionality Reduction · Geometric Methods in Data Analysis · Riemannian Geometry

Education

M.Sc. in Geometry (Topology)

University of Mohaghegh Ardabili

2022 – 2025 (expected / completed)

- GPA: **16.5 / 20**
 - Thesis: *The Mathematical Analysis of Manifold Learning Algorithms*
 - Focus areas:
 - Differential Geometry
 - Topology
 - Spectral methods in data analysis
 - Geometric structure of manifold learning algorithms
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B.Sc. in Educational Mathematics

Farhangian University of Tabriz

2017 – 2021

- GPA: **16.5 / 20**

- **Strong foundation in:** Mathematical Problem Solving, Mathematical Reasoning, Mathematical Thinking, Learning and Teaching Mathematics, and Educational Psychology , Linear Algebra, Real Analysis, Differential Equations
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Research Experience

Graduate Researcher – Manifold Learning & Geometric Machine Learning

Supervisor: Dr. Nikrouz Heydari

2024 – Present

- Developing theoretical extensions of **manifold learning algorithms** with emphasis on geometric structure preservation
 - Studying connections between:
 - spectral embeddings
 - graph Laplacians
 - Riemannian geometry
 - Investigating mathematical principles underlying dimensionality reduction techniques such as:
 - Isomap
 - Locally Linear Embedding (LLE)
 - Laplacian Eigenmaps
 - Diffusion Maps
 - Implementing numerical experiments in Python to validate theoretical findings
 - Contributing to ongoing research on **advanced dimensionality reduction frameworks**
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Publications & Conference Papers

A Comparative Mathematical Study of Manifold Learning Algorithms with Emphasis on Riemannian Manifold Learning

- Presented at international conference
- IEEE-indexed

<https://ieeexplore.ieee.org/document/11567405>

- Contribution:
 - Theoretical comparison of manifold learning methods
 - Emphasis on Riemannian geometric interpretation of embeddings
 - Analysis of spectral and local linear approximations in nonlinear dimensionality reduction
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Research Presentations

- Oral presentation: *Manifold Learning Algorithms and Riemannian Geometry*

2nd International Biennial Conference on Advances in Artificial Intelligence and Data Science (2026)

Teaching Experience

Mathematics Instructor (Secondary & High School Level)

2021 – Present (part-time)

- Teaching mathematics to high school and middle school students
 - Translating abstract mathematical concepts into intuitive explanations
 - Strengthening communication and pedagogical skills
 - Experience in structured lesson planning and student mentoring
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Programming & Computational Skills

- **Python** (NumPy, SciPy, basic ML implementations) (intermediate level)
 - Mathematical modeling and numerical simulation
 - Algorithm implementation for manifold learning methods
 - Basic experience with data analysis pipelines
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Current Research Direction

- Geometric analysis of dimensionality reduction algorithms
 - Extension of manifold learning frameworks using differential geometry
 - Investigation of connections between:
 - spectral graph theory
 - Riemannian geometry
 - representation learning
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Languages

- Persian — Native
 - English — Professional working proficiency (approx. IELTS 6.5–7.0 level)
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Research Objective

To investigate how artificial intelligence influences scientific problem-solving, reasoning, and knowledge production, with particular interest in AI-augmented scientific discovery and human–AI collaboration. Building on my background in mathematics education, mathematical problem-solving, geometry and topology, and machine learning, I aim to explore how AI systems interact with human reasoning and creativity in scientific research, and how the integration of AI into scientific practice may transform the processes through which researchers formulate, approach, and solve complex scientific problems.

References

Available upon request

Dr. Nikrouz Heydari (Supervisor)

E-mail: nikrooz.niky@gmail.com

Dr. Dariush Latifi

E-mail: latifi@uma.ac.ir